

pt 0 - In part 4, your goal is to investigate differences in gene expression between the electric organ and skeletal muscle. Write the null hypothesis for this goal below.

H_0 = Gene expression in the electric organ and skeletal muscle are the same.

```
library(edgeR)
```

```
## Loading required package: limma
```

```
library(limma)
library(Glimma)
library(ggplot2)
library(RColorBrewer)
library(DESeq2)
```

```
## Loading required package: S4Vectors
```

```
## Loading required package: stats4
```

```
## Loading required package: BiocGenerics
```

```
## Loading required package: generics
```

```
##
## Attaching package: 'generics'
```

```
## The following objects are masked from 'package:base':
##
##   as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
##   setequal, union
```

```
##
## Attaching package: 'BiocGenerics'
```

```
## The following object is masked from 'package:limma':
##
##   plotMA
```

```
## The following objects are masked from 'package:stats':
##
##   IQR, mad, sd, var, xtabs
```

```
## The following objects are masked from 'package:base':
##
##   anyDuplicated, aperm, append, as.data.frame, basename, cbind,
##   colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
##   get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
##   order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
##   rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
##   unsplit, which.max, which.min
```

```

##
## Attaching package: 'S4Vectors'

## The following object is masked from 'package:utils':
##
##     findMatches

## The following objects are masked from 'package:base':
##
##     expand.grid, I, unname

## Loading required package: IRanges

##
## Attaching package: 'IRanges'

## The following object is masked from 'package:grDevices':
##
##     windows

## Loading required package: GenomicRanges

## Loading required package: Seqinfo

## Loading required package: SummarizedExperiment

## Loading required package: MatrixGenerics

## Loading required package: matrixStats

##
## Attaching package: 'MatrixGenerics'

## The following objects are masked from 'package:matrixStats':
##
##     colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
##     colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
##     colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
##     colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
##     colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
##     colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
##     colWeightedMeans, colWeightedMedians, colWeightedSds,
##     colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
##     rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
##     rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
##     rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
##     rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
##     rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
##     rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
##     rowWeightedSds, rowWeightedVars

```

```

## Loading required package: Biobase

## Welcome to Bioconductor
##
##     Vignettes contain introductory material; view with
##     'browseVignettes()'. To cite Bioconductor, see
##     'citation("Biobase")', and for packages 'citation("pkgname)".

##
## Attaching package: 'Biobase'

## The following object is masked from 'package:MatrixGenerics':
##
##     rowMedians

## The following objects are masked from 'package:matrixStats':
##
##     anyMissing, rowMedians

library(dplyr)

##
## Attaching package: 'dplyr'

## The following object is masked from 'package:Biobase':
##
##     combine

## The following object is masked from 'package:matrixStats':
##
##     count

## The following objects are masked from 'package:GenomicRanges':
##
##     intersect, setdiff, union

## The following object is masked from 'package:Seqinfo':
##
##     intersect

## The following objects are masked from 'package:IRanges':
##
##     collapse, desc, intersect, setdiff, slice, union

## The following objects are masked from 'package:S4Vectors':
##
##     first, intersect, rename, setdiff, setequal, union

## The following objects are masked from 'package:BiocGenerics':
##
##     combine, intersect, setdiff, setequal, union

```

```

## The following object is masked from 'package:generics':
##
##     explain

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union

library(pheatmap)
library(clusterProfiler)

##

## clusterProfiler v4.20.0 Learn more at https://yulab-smu.top/contribution-knowledge-mining/
##
## Please cite:
##
## T Wu, E Hu, S Xu, M Chen, P Guo, Z Dai, T Feng, L Zhou, W Tang, L Zhan,
## X Fu, S Liu, X Bo, and G Yu. clusterProfiler 4.0: A universal
## enrichment tool for interpreting omics data. The Innovation. 2021,
## 2(3):100141

##
## Attaching package: 'clusterProfiler'

## The following object is masked from 'package:IRanges':
##
##     slice

## The following object is masked from 'package:S4Vectors':
##
##     rename

## The following object is masked from 'package:stats':
##
##     filter

library(KEGGREST)
library(enrichplot)

## enrichplot v1.32.0 Learn more at https://yulab-smu.top/contribution-knowledge-mining/
##
## Please cite:
##
## Guangchuang Yu, Li-Gen Wang, Yanyan Han and Qing-Yu He.
## clusterProfiler: an R package for comparing biological themes among
## gene clusters. OMICS: A Journal of Integrative Biology. 2012,
## 16(5):284-287

```

```
library(apeglm)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v forcats 1.0.1      v stringr 1.6.0
## v lubridate 1.9.5    v tibble 3.3.1
## v purrr 1.2.2       v tidyr 1.3.2
## v readr 2.2.0
```

```
## -- Conflicts ----- tidyverse_conflicts() --
## x lubridate::%within%() masks IRanges::%within%()
## x dplyr::collapse() masks IRanges::collapse()
## x dplyr::combine() masks Biobase::combine(), BiocGenerics::combine()
## x dplyr::count() masks matrixStats::count()
## x dplyr::desc() masks IRanges::desc()
## x tidyr::expand() masks S4Vectors::expand()
## x clusterProfiler::filter() masks dplyr::filter(), stats::filter()
## x dplyr::first() masks S4Vectors::first()
## x dplyr::lag() masks stats::lag()
## x BiocGenerics::Position() masks ggplot2::Position(), base::Position()
## x purrr::reduce() masks GenomicRanges::reduce(), IRanges::reduce()
## x clusterProfiler::rename() masks dplyr::rename(), S4Vectors::rename()
## x lubridate::second() masks S4Vectors::second()
## x lubridate::second<-() masks S4Vectors::second<-()
## x purrr::simplify() masks clusterProfiler::simplify()
## x clusterProfiler::slice() masks dplyr::slice(), IRanges::slice()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library("PoiClaClu")
library("glmpca")
library(ashr)
```

```
##
## Attaching package: 'ashr'
##
## The following object is masked from 'package:generics':
##
##      prune
```

pt 1 - Load in metadata and counts I decided to look at Species and Tissue since species is directly related to electric organ discharge, and tissue will allow us to study differences in EO and SM. Adding in the interaction term between Tissue and Species allows for further analysis of how they are interacting.

```
metadata = read.csv("metadata/Project2_RNAseq_metadata_campy.csv") %>%
  filter(AgeSize=="adult") %>%
  filter(Sample!="com15")
```

```
Campy = DESeqDataSetFromHTSeqCount(metadata,directory="counts_tar",design=~Species+Tissue+Species:Tissue)
```

```
## Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
## design formula are characters, converting to factors
```

pt 2 - filter reads I filtered reads by counts per million in order to filter reads that are lowly expressed. CPM takes sequencing depth into account and is therefore more accurate than just counts.

```
CPM = cpm(Campy)
campy_thresh = CPM > 0.5
retain = rowSums(campy_thresh) >= 2
Campy = Campy[retain,]
```

pt 3 - DEseq

```
Campy = DESeq(Campy)
```

```
## estimating size factors
```

```
## estimating dispersions
```

```
## gene-wise dispersion estimates
```

```
## mean-dispersion relationship
```

```
## final dispersion estimates
```

```
## fitting model and testing
```

```
Campy_results = results(Campy)
```

```
resultsNames(Campy)
```

```
## [1] "Intercept"
## [2] "Species_CampylomormyruscompressirostrisxCampylomormyrusrhynchophorus_vs_Campylomormyruscompressirostris"
## [3] "Species_CampylomormyruscompressirostrisxCampylomormyrustshokwe_vs_Campylomormyruscompressirostris"
## [4] "Species_Campylomormyrusrhynchophorus_vs_Campylomormyruscompressirostris"
## [5] "Species_Campylomormyrustshokwe_vs_Campylomormyruscompressirostris"
## [6] "Tissue_SM_vs_EO"
## [7] "SpeciesCampylomormyruscompressirostrisxCampylomormyrusrhynchophorus.TissueSM"
## [8] "SpeciesCampylomormyruscompressirostrisxCampylomormyrustshokwe.TissueSM"
## [9] "SpeciesCampylomormyrusrhynchophorus.TissueSM"
## [10] "SpeciesCampylomormyrustshokwe.TissueSM"
```

```
Campy_results = lfcShrink(Campy, coef="Tissue_SM_vs_EO", type="apeglm")
```

```
## using 'apeglm' for LFC shrinkage. If used in published research, please cite:
##   Zhu, A., Ibrahim, J.G., Love, M.I. (2018) Heavy-tailed prior distributions for
##   sequence count data: removing the noise and preserving large differences.
##   Bioinformatics. https://doi.org/10.1093/bioinformatics/bty895
```

```
## The difference in expression was quantified for each gene
```

```
### adjusted p-value is important here because we completed many statistical tests and need to adjust d
Campy_results[1:4,]
```

```
## log2 fold change (MAP): Tissue SM vs EO
## Wald test p-value: Tissue SM vs EO
## DataFrame with 4 rows and 5 columns
##
```

	baseMean	log2FoldChange
##	<numeric>	<numeric>
## augustus_masked-ptg0000251-processed-gene-5.22-mRNA-1	9.64009	-0.00671945
## augustus_masked-ptg0000671-processed-gene-3.69-mRNA-1	64.47568	-0.56452496
## augustus_masked-ptg0000841-processed-gene-4.62-mRNA-1	7.80369	0.39035263
## augustus_masked-ptg0001621-processed-gene-7.0-mRNA-1	16.70013	-0.25559553

```
##
```

	lfcSE	pvalue
##	<numeric>	<numeric>
## augustus_masked-ptg0000251-processed-gene-5.22-mRNA-1	0.733663	0.998511
## augustus_masked-ptg0000671-processed-gene-3.69-mRNA-1	0.519568	0.197060
## augustus_masked-ptg0000841-processed-gene-4.62-mRNA-1	0.930807	0.419083
## augustus_masked-ptg0001621-processed-gene-7.0-mRNA-1	0.643070	0.611532

```
##
```

	padj
##	<numeric>
## augustus_masked-ptg0000251-processed-gene-5.22-mRNA-1	0.998801
## augustus_masked-ptg0000671-processed-gene-3.69-mRNA-1	0.274504
## augustus_masked-ptg0000841-processed-gene-4.62-mRNA-1	0.510717
## augustus_masked-ptg0001621-processed-gene-7.0-mRNA-1	0.691025

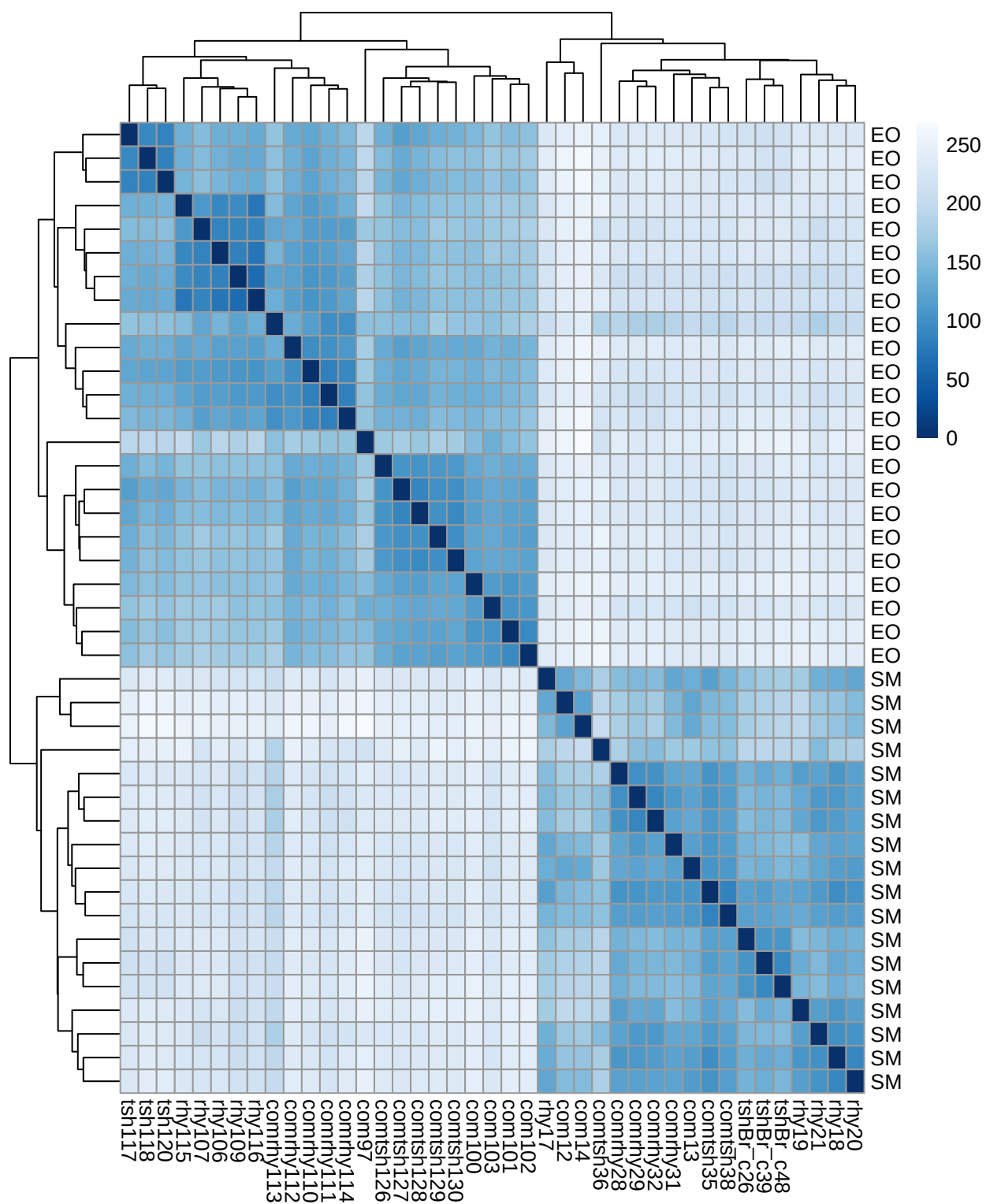
pt 4 - Perform rlog transformation rlog transforms the data to log2 form with respect to library size by minimizing differences between samples with low counts.

```
rCampy<-rlog(Campy)
```

```
## rlog() may take a few minutes with 30 or more samples,
## vst() is a much faster transformation
```

pt 5 - Heatmap First Heatmap – on rlog data to ensure equal contribution. Compares how similar samples are to each other. There was an outlier of SM within the EO section but it was removed in pt 7. Second Heatmap – Poisson, acts on the raw data. Does the same as the above heatmap. Both heatmaps show the same patterns, just with different numbers.

```
sampleDists = dist(t(assay(rCampy)))
sampleDistMatrix = as.matrix( sampleDists )
rownames(sampleDistMatrix) = rCampy$Tissue
colors = colorRampPalette( rev(brewer.pal(9, "Blues")) )(255)
pheatmap(sampleDistMatrix,
          clustering_distance_rows = sampleDists,
          clustering_distance_cols = sampleDists,
          col = colors)
```



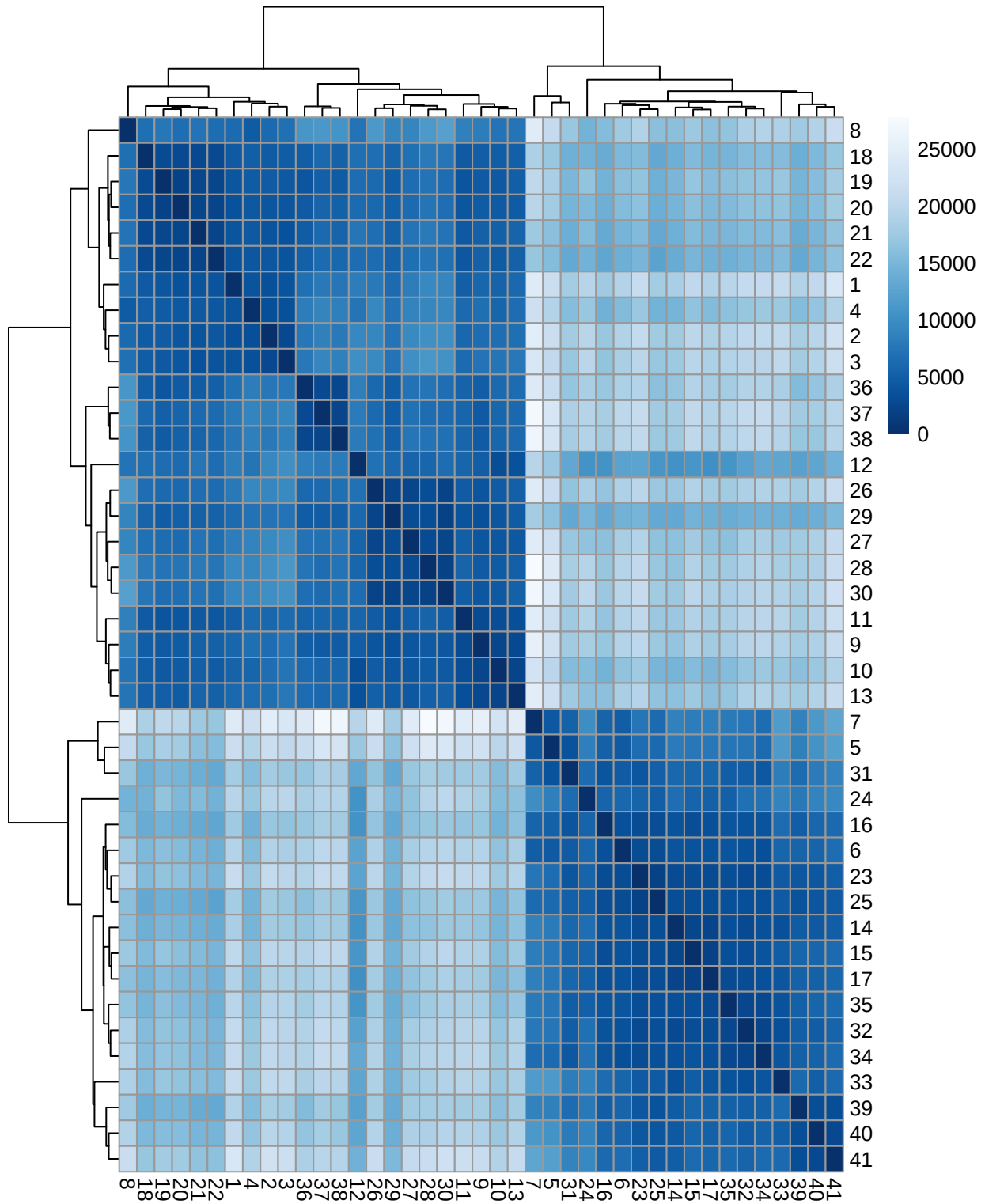
Poisson distance heatmap (takes unnormalized)

```
poisCampy <- PoissonDistance(t(counts(Campy)))
samplePoisDistMatrix <- as.matrix(poisCampy$dd )
rownames(sampleDistMatrix) = rCampy$Tissue
pheatmap(samplePoisDistMatrix,
```

```

clustering_distance_rows = poisCampy$dd,
clustering_distance_cols = poisCampy$dd,
col = colors)

```



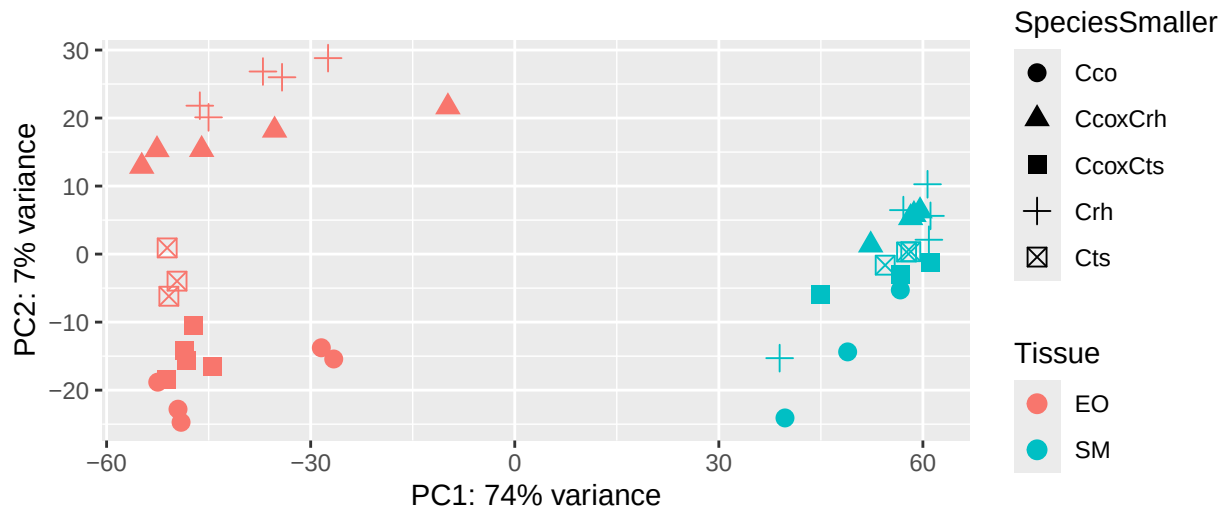
pt 6 - Examine similarity with PCA, gen PCA, MDS All plots clearly show separation based primarily

on tissue, with a secondary grouping based on species (related to EO discharge duration) within the EO cluster. Similarly to the heatmaps, the first PCA plot works on normalized data while the generalized one works on raw data. All plots showed an outlier, which was removed in part 7. MDS is less focused on capturing variance and instead focuses on preserving pairwise distances.

```
pcaData <- plotPCA(rCampy, intgroup=c("Tissue", "Species"), returnData=TRUE)
```

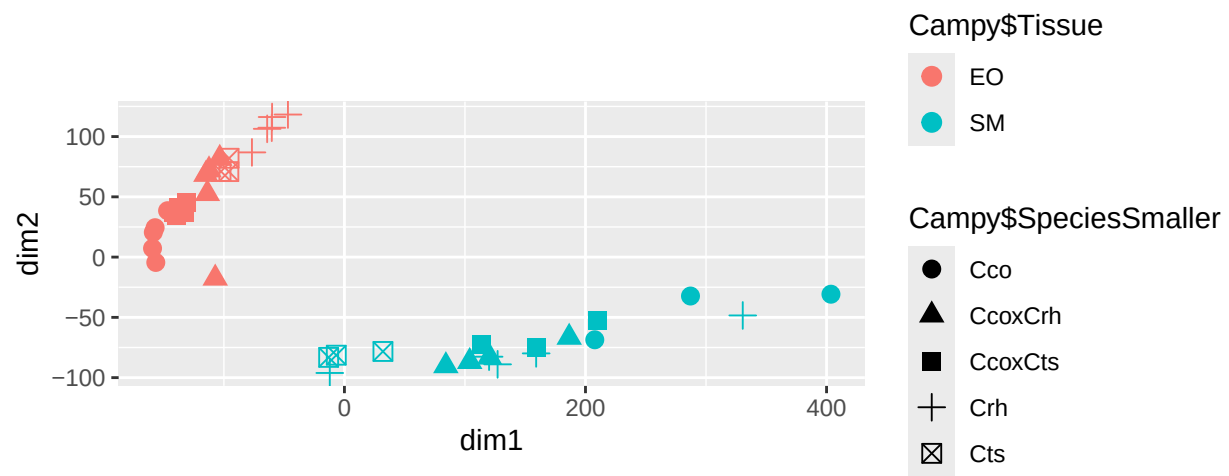
```
## using ntop=500 top features by variance
```

```
percentVar <- round(100 * attr(pcaData, "percentVar"))
ggplot(pcaData, aes(PC1, PC2, color=Tissue, shape=SpeciesSmaller)) +
  geom_point(size=3) +
  xlab(paste0("PC1: ", percentVar[1], "% variance")) +
  ylab(paste0("PC2: ", percentVar[2], "% variance")) +
  coord_fixed()
```



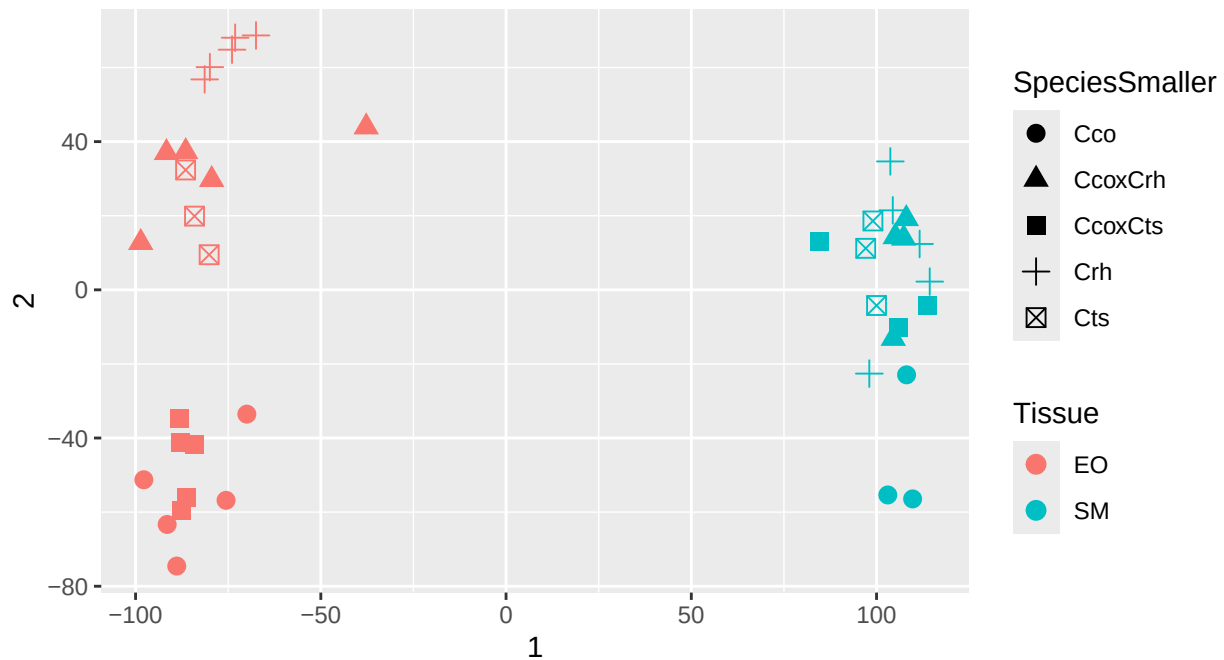
Generalized PCA

```
gpca <- glmpca(counts(Campy), L=2)
gpca.dat <- gpca$factores
ggplot(gpca.dat, aes(x = dim1, y = dim2, color = Campy$Tissue, shape = Campy$SpeciesSmaller)) +
  geom_point(size = 3) + coord_fixed()
```



MDS plot

```
mds <- as.data.frame(colData(rCampy)) %>%
  cbind(cmdscale(sampleDistMatrix))
ggplot(mds, aes(x = `1`, y = `2`, color = Tissue, shape = SpeciesSmaller)) +
  geom_point(size = 3) + coord_fixed()
```



Pt 7 - Filter outliers There was an outlier found in both the heatmaps and the PCA/MDS plots. One of the SM data points clustered with the EO.

Pt 8 - results table

```
resultsNames(Campy)
```

```
## [1] "Intercept"
## [2] "Species_CampylomoryruscompressirostrisxCampylomoryrusrhynchophorus_vs_Campylomoryruscompressirostris"
## [3] "Species_CampylomoryruscompressirostrisxCampylomoryrustshokwe_vs_Campylomoryruscompressirostris"
## [4] "Species_Campylomoryrusrhynchophorus_vs_Campylomoryruscompressirostris"
## [5] "Species_Campylomoryrustshokwe_vs_Campylomoryruscompressirostris"
## [6] "Tissue_SM_vs_EO"
## [7] "SpeciesCampylomoryruscompressirostrisxCampylomoryrusrhynchophorus.TissueSM"
## [8] "SpeciesCampylomoryruscompressirostrisxCampylomoryrustshokwe.TissueSM"
## [9] "SpeciesCampylomoryrusrhynchophorus.TissueSM"
## [10] "SpeciesCampylomoryrustshokwe.TissueSM"
```

```
cco = data.frame(lfcShrink(Campy, coef="Tissue_SM_vs_EO", type="ashr")) %>%
  rownames_to_column(var = "gene") %>%
  select(gene, log2FoldChange, padj) %>%
  filter(abs(log2FoldChange) > 1, padj < 0.05) %>%
  rename("ccolog" = "log2FoldChange") %>%
  rename("ccopadj" = "padj")
```

using 'ashr' for LFC shrinkage. If used in published research, please cite:

```
## Stephens, M. (2016) False discovery rates: a new deal. Biostatistics, 18:2.
## https://doi.org/10.1093/biostatistics/kxw041
```

```
ccoxcrh = data.frame(lfcShrink(Campy, contrast=list(c("Tissue_SM_vs_EO", "SpeciesCampylomormyruscompressus")),
  rownames_to_column(var = "gene") %>%
  select(gene, log2FoldChange, padj) %>%
  filter(abs(log2FoldChange) > 1, padj < 0.05) %>%
  rename("ccoxcrhlog" = "log2FoldChange") %>%
  rename("ccoxcrhpadj" = "padj"))
```

```
## using 'ashr' for LFC shrinkage. If used in published research, please cite:
## Stephens, M. (2016) False discovery rates: a new deal. Biostatistics, 18:2.
## https://doi.org/10.1093/biostatistics/kxw041
```

```
ccoxcts = data.frame(lfcShrink(Campy, contrast=list(c("Tissue_SM_vs_EO", "SpeciesCampylomormyruscompressus")),
  rownames_to_column(var = "gene") %>%
  select(gene, log2FoldChange, padj) %>%
  filter(abs(log2FoldChange) > 1, padj < 0.05) %>%
  rename("ccoxctslog" = "log2FoldChange") %>%
  rename("ccoxctspadj" = "padj"))
```

```
## using 'ashr' for LFC shrinkage. If used in published research, please cite:
## Stephens, M. (2016) False discovery rates: a new deal. Biostatistics, 18:2.
## https://doi.org/10.1093/biostatistics/kxw041
```

```
crh = data.frame(lfcShrink(Campy, contrast=list(c("Tissue_SM_vs_EO", "SpeciesCampylomormyrusrhynchophorus")),
  rownames_to_column(var = "gene") %>%
  select(gene, log2FoldChange, padj) %>%
  filter(abs(log2FoldChange) > 1, padj < 0.05) %>%
  rename("crhlog" = "log2FoldChange") %>%
  rename("crhpadj" = "padj"))
```

```
## using 'ashr' for LFC shrinkage. If used in published research, please cite:
## Stephens, M. (2016) False discovery rates: a new deal. Biostatistics, 18:2.
## https://doi.org/10.1093/biostatistics/kxw041
```

```
cts = data.frame(lfcShrink(Campy, contrast=list(c("Tissue_SM_vs_EO", "SpeciesCampylomormyrustshokwe.Tissotshokwe")),
  rownames_to_column(var = "gene") %>%
  select(gene, log2FoldChange, padj) %>%
  filter(abs(log2FoldChange) > 1, padj < 0.05) %>%
  rename("ctslog" = "log2FoldChange") %>%
  rename("ctspadj" = "padj"))
```

```
## using 'ashr' for LFC shrinkage. If used in published research, please cite:
## Stephens, M. (2016) False discovery rates: a new deal. Biostatistics, 18:2.
## https://doi.org/10.1093/biostatistics/kxw041
```

```
#turn row names into a column = rownames_to_column
#filter where p value is below .05 and logfold change >1
#rename log and p and select for columns u want
```

```

#repeat with the other species one and join the data together -- innerjoin()
Campy_DEG = inner_join(cco, ccoxcrh, by="gene") %>%
  inner_join(ccoxcts, by="gene") %>%
  inner_join(crh, by="gene") %>%
  inner_join(cts, by="gene")

#group_by and summarize
mean_logs = Campy_DEG %>%
  select(contains("log")) %>%
  rowMeans()
mean_padj = Campy_DEG %>%
  select(contains("padj")) %>%
  rowMeans()

Campy_DEG = Campy_DEG %>%
  mutate("mean_logs" = mean_logs) %>%
  mutate("mean_padj" = mean_padj)

#volcano, multiply all values by -1

```

Volcano plot

```

# Compute significance, with a maximum of 320 for the p-values set to 0 due to limitation of computation
Campy_DEG$sig <- -log10(Campy_DEG$mean_padj)
# View(resultDESeq2[is.na(resultDESeq2$pvalue),])

# Select genes with a defined p-value (DESeq2 assigns NA to some genes)
genes.to.plot <- !is.na(Campy_DEG$mean_padj)
# sum(genes.to.plot)
range(Campy_DEG[genes.to.plot, "log2FoldChange"])

```

```

## Warning in min(x, na.rm = na.rm): no non-missing arguments to min; returning
## Inf

```

```

## Warning in max(x, na.rm = na.rm): no non-missing arguments to max; returning
## -Inf

```

```

## [1] Inf -Inf

```

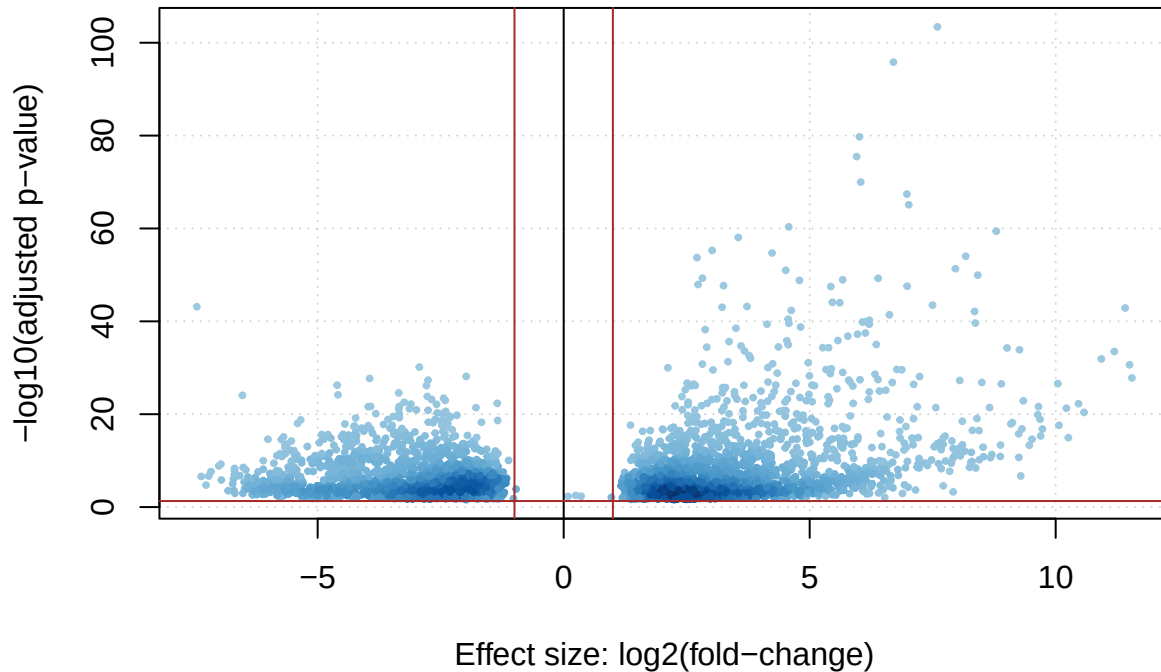
```

## Volcano plot of adjusted p-values
cols <- densCols(-Campy_DEG$mean_logs, Campy_DEG$sig)
cols[-Campy_DEG$mean_padj ==0] <- "purple"
Campy_DEG$pch <- 19
Campy_DEG$pch[-Campy_DEG$mean_padj ==0] <- 6
plot(-Campy_DEG$mean_logs,
     Campy_DEG$sig,
     col=cols, panel.first=grid(),
     main="Volcano plot, right is up in EO, left is up in SM",
     xlab="Effect size: log2(fold-change)",
     ylab="-log10(adjusted p-value)",
     pch=Campy_DEG$pch, cex=0.4)
abline(v=0)

```

```
abline(v=c(-1,1), col="brown")
abline(h=-log10(0.05), col="brown")
```

Volcano plot, right is up in EO, left is up in SM



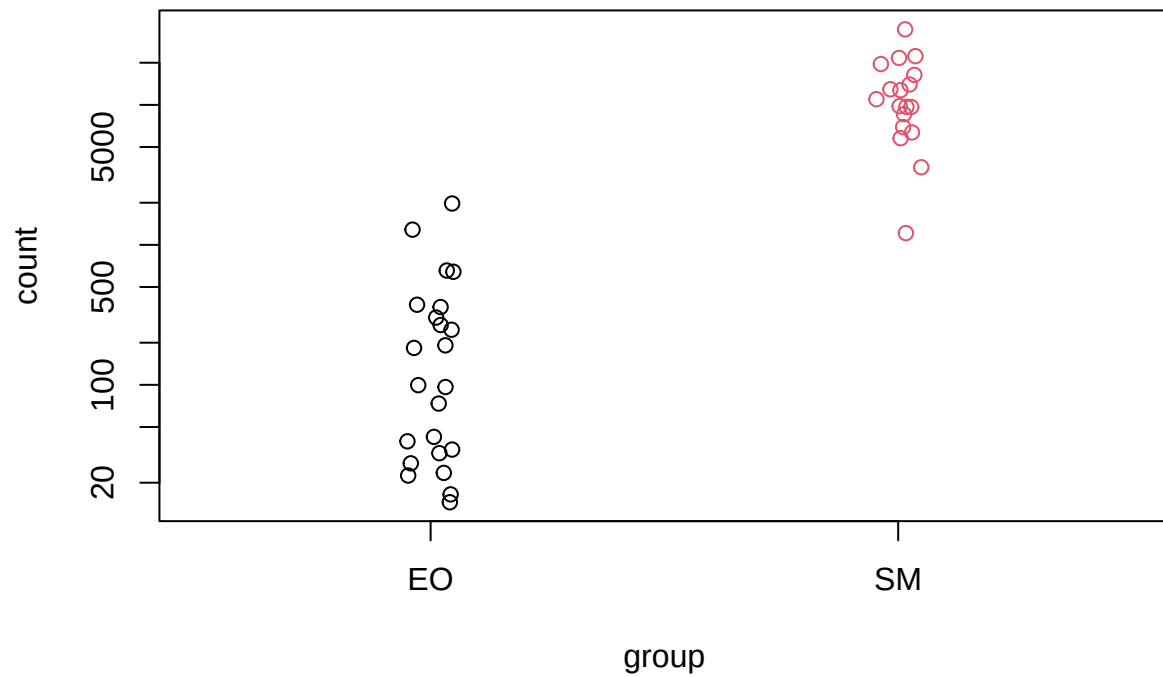
```
#right = EO
```

Pt 9 - Visualize individual candidates

```
#top 5 lowest padj
top_5 = Campy_DEG %>%
  arrange(mean_padj) %>%
  head(n=5) %>%
  select(gene)

#gene with highest log2fc
top_log = Campy_DEG %>%
  arrange(-mean_logs) %>%
  head(n=1)
plotCounts(Campy, gene="snap_masked-ptg0011431-processed-gene-2.56-mRNA-1", intgroup="Tissue", normalize=TRUE)
```

snap_masked-ptg001143l-processed-gene-2.56-mRNA-1



```
#top EO (lowest log2fc)
top_EO = Campy_DEG %>%
  arrange(mean_logs) %>%
  head(n=1)
plotCounts(Campy, gene="maker-ptg000521l-snap-gene-0.25-mRNA-1", intgroup="Tissue", normalized=FALSE, c
```

maker-ptg000521l-snap-gene-0.25-mRNA-1

